Class fairness in model-heterogeneous federated learning

A measurement study, a diagnostic ladder, and a research roadmap

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Repository: github.com/Paul4805/phofl-class-fairness

Artifacts: phofl/analysis, phofl/paper, phofl_method, phofl_half

The short version

Client-FM - the usual standard deviation of per-client accuracy - hides a **0.39 +/- 0.08** best-to-worst class-recall spread in PHP-FL/linear over three seeds.

The single-seed 44.3-point figure was a high draw; PHP-FL/linear spans 0.316-0.468 across seeds.

FedAPEN/linear is consistently worse: **0.60 +/- 0.09**, 6/6 per-seed comparisons, so the observation is field-wide across the two MH-FL methods tested.

A four-rung diagnostic ladder (rehearsal, confusion loss, worst-class aggregation, per-class distillation) reduced the gap by at most **7.9%** over one seed.

Step 2 is the clearest mechanism: dog-to-cat errors fell 23%, but cat recall fell 16%, and the combined dog+cat error rose 0.657 -> 0.683.

class_FM is a **reporting metric**, not a training objective. No method contribution is claimed.

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Executive summary

This report documents a measurement project that started from the PHP-FL paper and ended with a two-axis fairness critique plus a negative-result diagnostic ladder. The central claim is simple: a single client-level dispersion number is not a complete fairness report for model-heterogeneous federated learning (MH-FL). A model can look well balanced across clients while leaving a large semantic class spread.

On the half-scale CIFAR-10 setup, PHP-FL/linear has client-FM 0.0860 and a class gap of **0.39 +/- 0.08** over three seeds, with seed 0 at 0.468. The single best client-FM cell (FedAPEN/linear, FM=0.0779) is the worst on the class axis at 0.60 +/- 0.09. Because FedAPEN crosses the pre-registered 30-point scope threshold in every seed and is worse than PHP-FL in 6/6 per-seed comparisons, the finding is field-wide across the two MH-FL methods in this study, but it is not a universal claim about all FL.

The eight cells also order differently on the two axes: the worst-gap cell is also the best-client-FM cell in two of three seeds. With three seeds this is a replicated ordering observation, not a rank-correlation claim.

The diagnostic ladder was designed to test whether the gap is reducible. Four standard interventions were run: class-balanced rehearsal, confusion-targeted loss, worst-class-weighted aggregation, and per-class distillation. The largest reduction was 7.9%, from rehearsal. Step 2 is the most informative: it moved dog-to-cat errors in the intended direction but shifted the error out of the pair, into bird, leaving the combined dog+cat error slightly higher. The full 10x10 matrices and a centralized probe show this is an error-allocation and representation problem, not an information floor at 32x32, and argue for class_FM as a reporting metric rather than an optimization target.

1. The PHP-FL paper in brief

PHP-FL addresses fairness in model-heterogeneous federated learning when clients have different participation probabilities. The setting is deliberately heterogeneous: four CNN architectures are assigned round-robin to 20 clients, so there is no single global model that can simply be averaged across all clients. The paper's fairness target is client-level: reduce the dispersion of per-client test accuracy under participation inconsistency.

1.1 Method components

- **Private and auxiliary models.** Each client has a private model w_i and a local copy g_i of a shared auxiliary model. The server maintains the global auxiliary model G. The private model carries personalization; the auxiliary model carries shared knowledge.
- **DEAL: dual-end alignment.** The local objective combines cross-entropy with two alignment terms: an MMD term between the private and shared backbone features, and a KL term between their logits. An ensemble loss mixes the two predictions. The intent is to align representations and outputs at both ends while preserving personalization.
- **ISPU: individual-specific parameter update.** Each client computes a participation ratio alpha from its cumulative participation count N_i(t). The top-alpha parameters of its auxiliary model, ranked by L1 norm, are overwritten with the corresponding global parameters. In the repository the schedule is `alpha = tau * sigmoid(k, (N_i(t)+1)/(t+1))` with `sigmoid(k,x)=1/(1+exp(k(x-0.5)))`. Rare clients therefore receive a larger alpha and more global knowledge; frequent clients receive less.
- **Server aggregation.** Active clients upload their auxiliary models; the server takes a plain mean over active clients; active clients then apply their ISPU masks. The private models are not aggregated.
- **Evaluation.** AM is the mean per-client accuracy; FM is the standard deviation of per-client accuracy. Participation patterns are uniform, normal(mu, sigma), and a shuffled linear/arithmetic schedule.

1.2 What the paper claims

The paper claims that the combination of dual-end alignment and individual-specific updates improves client fairness under participation-probability inconsistency in heterogeneous environments. The primary evidence is the FM/AM table on CIFAR-10 and related datasets. The method is evaluated as a client-fairness method; class-level recall is not part of the reported metric.

1.3 What this project tests

This project does not re-derive or replace PHP-FL. It asks a complementary measurement question: if a method improves client-FM, does it also improve the class axis? The repository's PHP-FL code is used as the reference implementation; the class-fairness analysis is layered on top of the same training and evaluation logs.

2. Project arc: what was planned, what was run

The project began with a much larger plan: reproduce the paper's CIFAR-10 normal-pattern baseline (AM 66.94, FM 8.07), then run aggregation, participation, cold-start, convergence, efficiency, membership-inference, and baseline-porting experiments. The scope was reduced before the full baseline matrix ran. As a result, this repository does **not** contain a successful reproduction of Table 2, and no claim of exact paper reproduction is made.

The work that was actually completed followed this arc:

- **Phase A/B/C (Fashion-MNIST smoke).** Fairness-metric recomputation, aggregation reweighting, and correlated-participation comparisons on 10 clients, 40 rounds, two backbones. These are diagnostic smoke runs, not CIFAR-10 claims.
- **Participation branch closed.** The C escalation was pre-registered, corrected with Holm, and repeated on the pinned stack. The linear-vs-entropy pattern effect did not survive correction, and the half-scale run failed the pre-registered thresholds. The participation hypothesis is closed and is not revived.
- **Class-fairness pivot.** The analysis moved to the two-axis object: client x class. Five CIFAR-10 half-scale runs were analyzed: PHP-FL linear/entropy, FedAPEN linear, and Standalone linear/entropy.
- **FedAPEN scope test.** FedAPEN/linear was added as the second MH-FL method. Its 52.9-point class gap promoted the finding to field-wide across the two methods tested.
- **Diagnostic ladder.** Four interventions were tested one at a time to see whether the class gap is reducible. None closed it; the maximum reduction was 7.9%.

The most important consequence of the arc is methodological: the project changed from trying to build a better method to measuring whether the reported fairness metric captures the failure it claims to address. That change is the paper.

3. Experimental setup and metric definitions

3.1 Half-scale CIFAR-10 setup

Item	Value
Dataset	CIFAR-10, 10 clients, Dirichlet(alpha=0.5)
Per-client data	2,500 train / 500 test
Training	60 rounds, 1 local epoch, batch 64 (32 for heavy backbones)
Models	Four CNNs: GoogLeNet, DenseNet-121, EfficientNet-B1, ResNet-18
Auxiliary	GoogLeNet, plain mean aggregation
Seed	0 (one seed)
Stack	torch 2.2.2+cu121, numpy 1.26.4, pinned
Evaluation	Full 1,000-sample CIFAR-10 test set per class

This is a compute-bounded half-scale setting: roughly 8x less training than the paper's full CIFAR-10 setting. Absolute AM/FM are not comparable to the paper. The linear-vs-entropy contrast and the class-axis geometry are the usable outputs.

3.2 Metrics

- **AM:** mean per-client accuracy. **FM:** population standard deviation of per-client accuracy. In the artifacts, `AM_std` is numerically FM, not a seed-level standard error.
- **Class gap:** best per-class recall minus worst per-class recall on the full test set. **class_std:** population standard deviation of the 10 per-class recalls.
- **Worst-class recall:** minimum per-class recall. This is the tail metric that a client-level average can hide.
- **Support filter:** per-(client, class) cells with fewer than 10 test samples are excluded from worst-cell and support-filtered calculations. Each run has 34/100 such cells (170 across five runs; 45 have zero samples). The global class headline uses the full 1,000 samples per class and is not affected.
- **class_FM:** standard deviation over classes of per-class mean recall. It is the class-axis analogue of client-FM and is proposed as a reporting metric, not a training objective.
- **AM-normalized gap:** class gap divided by AM. This asks whether a smaller raw gap is simply a lower-accuracy effect. The ceiling-normalized variant divides by 1 - worst-class recall.

4. First measurement phase: A/B/C

The first phase was a Fashion-MNIST smoke study: 10 clients, 40 rounds, 1 seed, GoogLeNet and ResNet-18, 1,000 train / 200 test per client. It was designed to test the measurement pipeline and the participation branch, not to make CIFAR-10 claims.

4.1 A: fairness is more than std

Metric	Plain PHP-FL baseline
AM	0.5175
FM	0.1607
min accuracy	0.2400
10th percentile	0.2985
CVaR10	0.2400
CV	0.3105
Jain	0.9120
Spearman rho(p, accuracy)	-0.0061

The original A narrative expected PHP-FL to have the lowest FM but not the best tail. In this smoke baseline the FM winner also won min/CVaR, and rho was not positive. The narrative was not confirmed on this run. This is a negative result about the baseline, not about the CIFAR-10 paper.

4.2 B: aggregation reweighting was a null result

Aggregation	AM	FM	min
plain	0.5175	0.1607	0.240
inv_p	0.5160	0.1628	0.260
inv_N	0.5105	0.1615	0.260

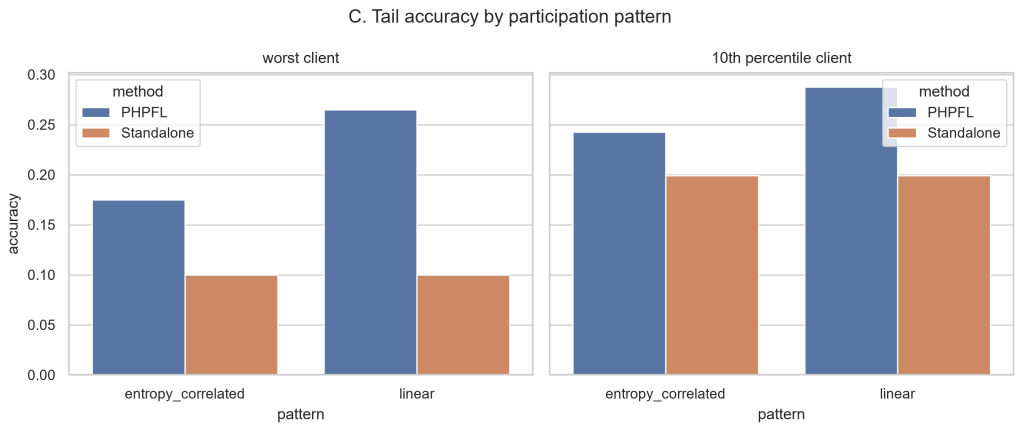
The tail minimum improved by about 2 points under inv_p and inv_N, but the paired Wilcoxon tests on client accuracy were null: plain vs inv_p p=0.922, plain vs inv_N p=0.469, inv_p vs inv_N p=0.570. The aggregation branch was written up as a null result and moved past.

4.3 C: correlated participation and the closed branch

Pattern	Method	AM	FM	min	rho(p,acc)
linear	PHPFL	0.5040	0.1439	0.265	-0.758
entropy	PHPFL	0.5120	0.1805	0.175	+0.721
linear	Standalone	0.4605	0.1856	0.100	-0.794

The entropy-correlated sampler raised mean accuracy slightly but increased dispersion and collapsed the tail. The raw linear-vs-entropy dispersion test was p=0.041; after Holm correction it became p=0.164 and lost significance. PHP-FL vs Standalone under entropy survived Holm (p=0.0098), but that is a method contrast, not a pattern effect. The pinned-stack smoke run preserved the sign but not the significance of the dispersion effect.

The C escalation then ran at CIFAR-10 10 clients / 60 rounds. It failed the pre-registered thresholds: FM increase -2.84% (GO required >25%), min accuracy +4.09 points (GO required a >5-point drop), rho difference 0.170 (GO required >1.0). C is closed. This report does not revive the participation hypothesis.



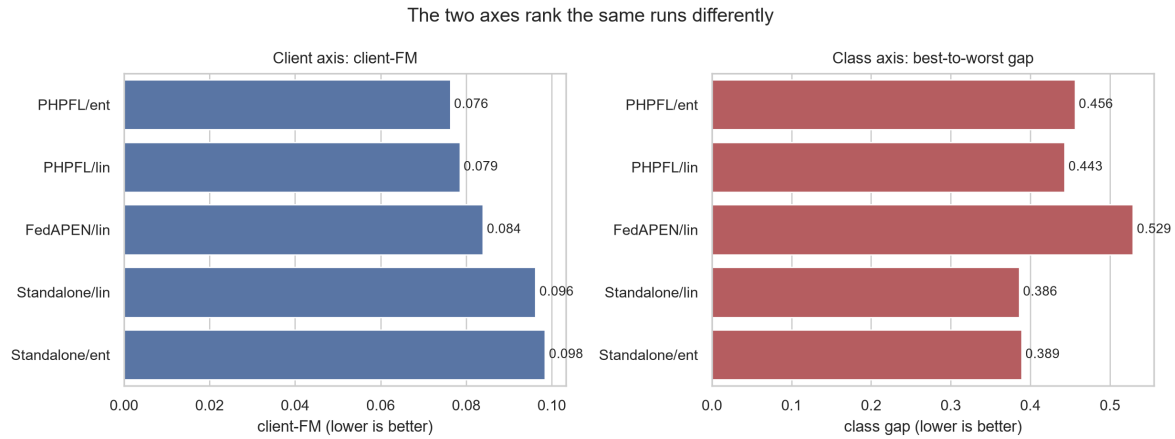
Fashion-MNIST smoke: entropy-correlated participation collapses the tail. This is a smoke diagnostic, not a CIFAR-10 result.

5. Two-axis fairness on CIFAR-10

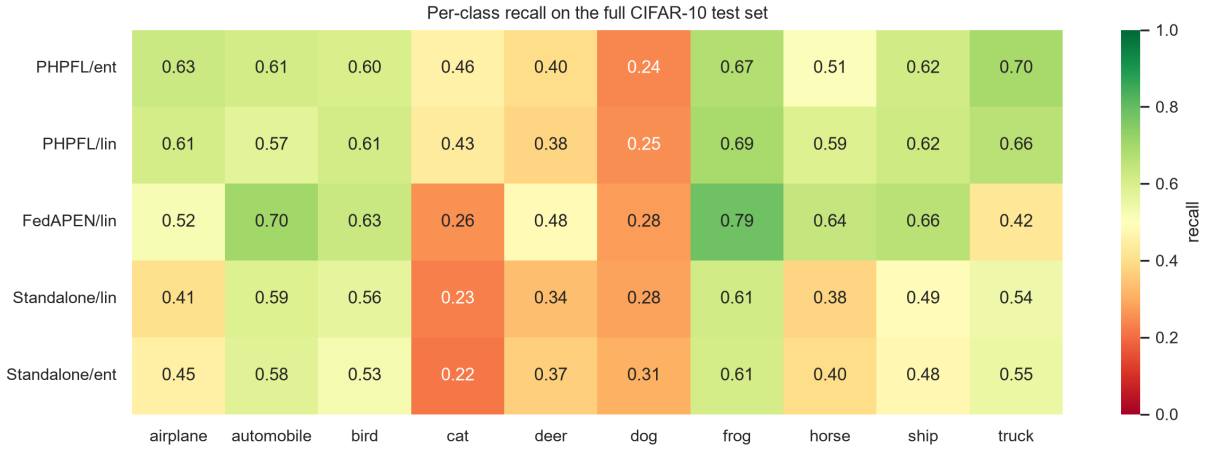
The central measurement result is the client-by-class matrix. Client-FM is the row-mean dispersion; class_std and worst-class recall are the column summaries. The table below is the original seed-0 measurement; subsection 5.1 gives the three-seed replication that supersedes its point estimates.

Run (seed 0)	AM	client-FM	class_std	class gap	worst class	worst recall
PHPFL/ent	0.6308	0.0763	0.1349	0.4560	dog	0.2400
PHPFL/lin	0.6320	0.0785	0.1334	0.4430	dog	0.2510
FedAPEN/lin	0.6209	0.0839	0.1698	0.5290	cat	0.2560
Standalone/lin	0.5429	0.0962	0.1281	0.3860	cat	0.2290
Standalone/ent	0.5467	0.0985	0.1196	0.3890	cat	0.2180

Seed 0 gave 44.3 points in PHP-FL/linear (frog 69.4%, dog 25.1%) and 52.9 for FedAPEN/linear. Those are single draws: re-running the same cells on seeds 1 and 2 moves PHP-FL/linear to 0.316 and 0.376 and FedAPEN/linear to 0.558 and 0.700. The failure is not a single-class artifact: dog, cat, deer, and horse form a broad weak region.



Client-FM and class gap at seed 0. The bars are ordered by client-FM on the left; the class axis on the right does not follow that order.



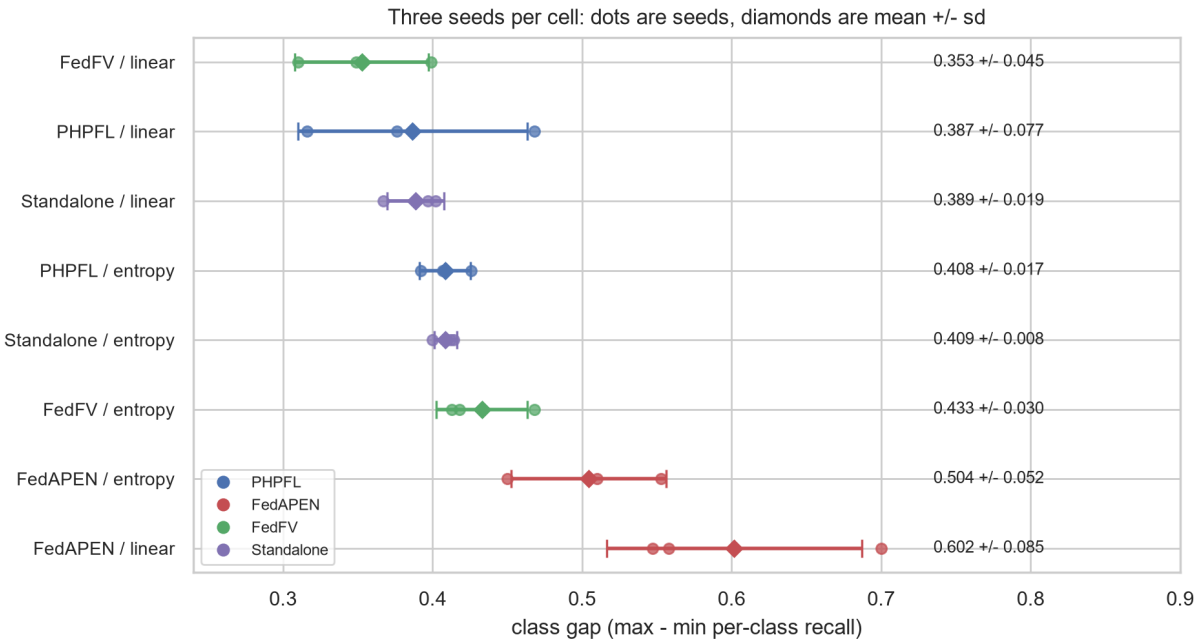
Per-class recall at seed 0. Rows are runs; columns are classes. The weak region is dog/cat/deer/horse, not one isolated cell.

5.1 Three-seed replication

Eight cells (PHP-FL, FedAPEN, FedFV, Standalone x linear/entropy) were re-run on seeds 0, 1 and 2 under one stack (torch 2.14.1), 24 runs in total. Class gap is mean +/- sample sd. The seed-0 runs were repeated on this stack so the three seeds are comparable; the pinned seed-0 artifacts differ by at most 0.005 AM and 0.012 client-FM.

Method / pattern	AM	client-FM	class gap	worst recall
PHPFL / entropy	0.6302 +/- 0.0041	0.0817 +/- 0.0032	0.4083 +/- 0.0170	0.278
PHPFL / linear	0.6286 +/- 0.0083	0.0860 +/- 0.0095	0.3867 +/- 0.0766	0.312
FedAPEN / entropy	0.6064 +/- 0.0075	0.0873 +/- 0.0109	0.5043 +/- 0.0517	0.276
FedAPEN / linear	0.6151 +/- 0.0030	0.0779 +/- 0.0137	0.6017 +/- 0.0853	0.207
FedFV / entropy	0.6262 +/- 0.0087	0.0847 +/- 0.0042	0.4330 +/- 0.0304	0.268
FedFV / linear	0.6272 +/- 0.0080	0.0874 +/- 0.0035	0.3527 +/- 0.0446	0.346
Standalone / entropy	0.5377 +/- 0.0060	0.0954 +/- 0.0100	0.4087 +/- 0.0076	0.212
Standalone / linear	0.5423 +/- 0.0014	0.0996 +/- 0.0095	0.3887 +/- 0.0189	0.215

Three consequences. First, seed variance is first-order: PHP-FL/linear spans 0.316-0.468 and FedAPEN/linear spans 0.547-0.700, so a single-seed gap is a draw, not a measurement. Second, PHP-FL has no class-gap advantage over Standalone (0.387 vs 0.389; per-seed differences +0.071, -0.051, -0.026 change sign), so the earlier AM-normalization argument is moot. Third, FedAPEN is worse than PHP-FL in 6/6 per-seed comparisons (penalty 6-32 points), which is the only ordering that survives reseeding. FedFV/linear has the best mean gap (0.353) and worst-class recall (0.346), but its 3.4-point edge over PHP-FL is inside the seed noise (3/3 sign, two-sided sign test p=0.25); it is reported as a directional observation, not a fix.



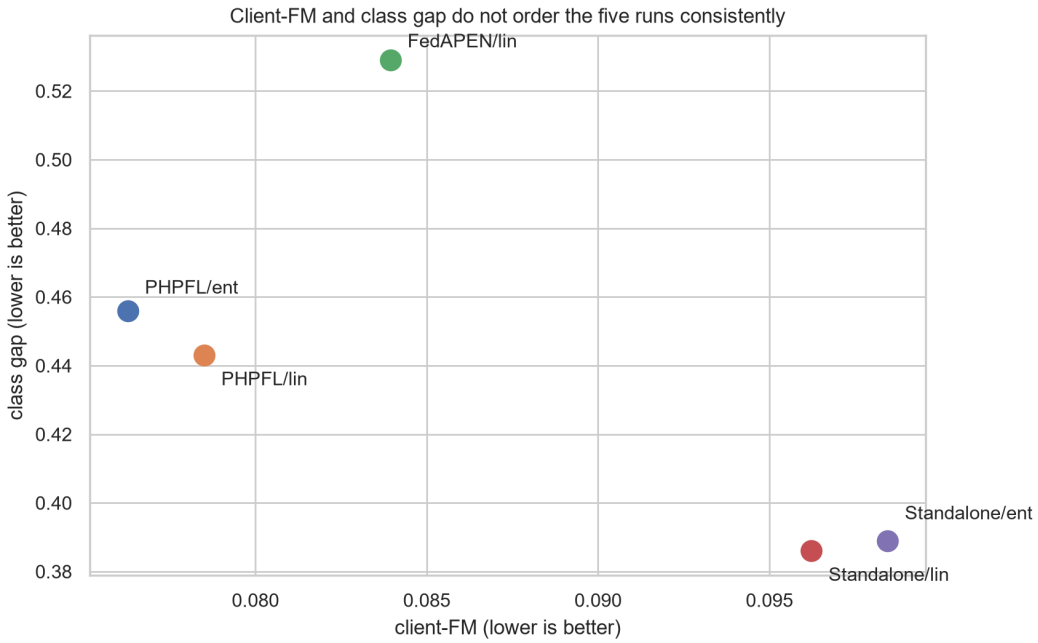
Per-seed class gap for every cell. Dots are seeds; the bar is the mean. PHP-FL and Standalone overlap; FedAPEN is separated upward.

5.2 Support filtering

Per-client class cells are sparse: 34/100 cells per run have fewer than 10 test samples, 170 cells across five runs, including 45 zero-sample cells. These are listed in `low_support_cells.csv` and excluded from the worst-cell calculation. The worst supported cell is still 0.0 (client 0, cat, 11 samples) in every run. It is sparse but above threshold and is not the headline. Computing class means only from supported cells gives a 51.3-point gap for PHP-FL/linear at seed 0, a 45.0-point gap for PHP-FL/entropy, and a 41.6-point gap for FedAPEN; the seed-0 gap was therefore conservative. The three-seed table is computed from the full test set.

5.3 Rankings disagree

At seed 0 the two metrics ordered the five original runs in opposite directions. With eight cells over three seeds the inversion is stronger: the worst-gap cell is FedAPEN/linear in all three seeds, and it is also the best-client-FM cell in seeds 0 and 2 (PHP-FL/linear takes best client-FM at seed 1). A leaderboard sorted by client-FM therefore selects, two times out of three, the run with the worst class gap.



Seed-0 runs in the client-FM / class-gap plane. The ordering is not monotone; the two axes disagree, and the reversal repeats across seeds.

5.4 The pattern effect is method-dependent

The seed-0 reading that class-std is almost invariant to the participation pattern does not replicate. Across three seeds the entropy-minus-linear gap is +0.015/+0.033/+0.012 for Standalone (worse 3/3), +0.069/+0.108/+0.064 for FedFV (worse 3/3), mixed for PHP-FL (-0.042/+0.076/+0.031), and -0.037/-0.108/-0.147 for FedAPEN (better 3/3). The class gap is large under both patterns, but the sampler interacts with the aggregation rule rather than acting as a neutral nuisance.

5.5 AM normalization is superseded

The earlier gap/AM adjustment assumed Standalone's raw gap was smaller (38.6 vs 44.3). With three seeds the raw gaps are tied (0.389 vs 0.387), so there is nothing to normalize away and we drop the Standalone class-fairness discussion entirely rather than report a normalized version of a difference that is not there. The PHP-FL vs FedAPEN inversion is AM-independent: FedAPEN still has the best client-FM and the worst class gap.

5.6 FedAPEN and the field-wide scope

FedAPEN/linear has a 0.60 +/- 0.09 class gap and FedAPEN/entropy 0.50 +/- 0.05; the seed-0 support-filtered gap was 41.6 points. The pre-registered rule was: >30 points promotes to field-wide, <20 narrows to PHP-FL, 20-30 is mixed. Every seed crosses 30 points and FedAPEN is worse than PHP-FL in all six method-pattern comparisons, so the claim is field-wide across the two MH-FL methods tested (PHP-FL and FedAPEN), not universal across all FL.

6. The diagnostic ladder

If a standard intervention could remove the class gap, the metric critique would be less interesting. The ladder tested four one-at-a-time interventions under the same half-scale configuration. The baseline is the existing PHP-FL/linear run.

Rung	Intervention	AM	FM	gap	reduction	dog	frog
0	PHP-FL/linear baseline	0.6320	0.0785	0.443	-	0.251	0.694
1	class-balanced rehearsal (50/class)	0.6099	0.0854	0.4080	+7.9%	0.3200	0.7280
2	confusion-targeted loss (top-3, lambda=0.1)	0.6346	0.0900	0.4500	-1.6%	0.2700	0.7200
3	worst-class-weighted aggregation (gamma=5)	0.6312	0.1002	0.4310	+2.7%	0.2920	0.7230
4	per-class distillation (T=1.0)	0.6322	0.0944	0.4240	+4.3%	0.2820	0.7060

No rung reached the 30% closure threshold. The maximum reduction was 7.9% from rehearsal. The ladder is therefore a negative result about reducibility, not a failed method paper.



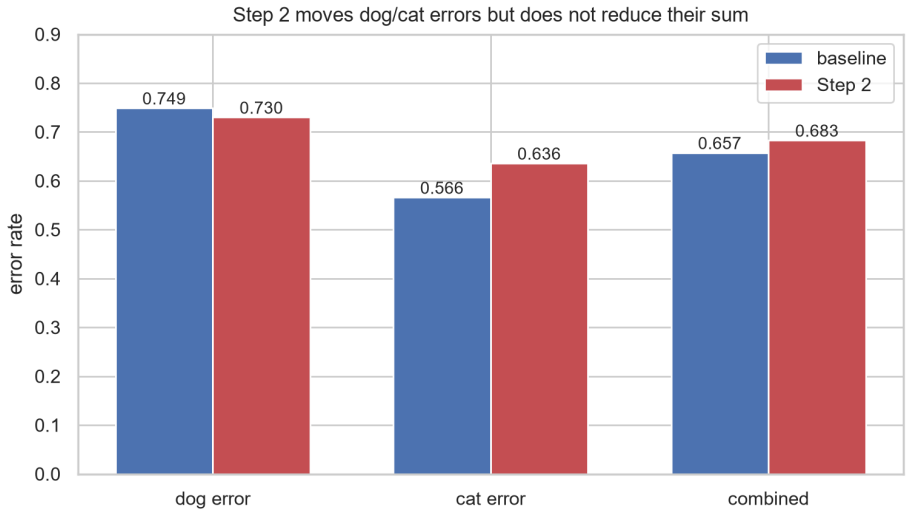
Ladder gaps and relative reduction. The 30% closure threshold is dashed; no rung approaches it.

6.1 Rehearsal: broad improvement, preserved gap

Step 1 stored 50 samples per class and mixed them 1:1 with fresh batches. Dog recall rose from 0.251 to 0.320 (+6.9 points) and frog from 0.694 to 0.728 (+3.4 points). The gap fell only from 0.443 to 0.408 (7.9%). Both ends improved, but the improvement was proportional rather than redistributive; the normalized gap barely moved.

6.2 Confusion loss: the zero-sum result

Step 2 added a margin penalty on each client's top-3 validation confusion pairs (lambda=0.1, margin 1.0). Dog-to-cat errors fell from 0.249 to 0.191 (-23%). But cat recall fell from 0.434 to 0.364 (-16%), and the combined dog+cat error rate rose from 0.657 to 0.683. The loss moved the decision boundary rather than removing the joint error. Section 6.5 shows where the mass went: into bird.



Step 2 shifts dog/cat errors: dog improves, cat worsens, the combined error does not fall.

6.3 Worst-class aggregation

Step 3 used weights exp(-gamma * recall_i(c_worst)) with gamma=5. The globally worst class was airplane early and dog later. The formula gives the largest weight to clients with low recall on the worst class, i.e. the clients least able to represent it. The gap fell only to 0.431 (2.7%); dog recall reached 0.292. Aggregation reweighting alone does not fix a representation problem.

6.4 Per-class distillation

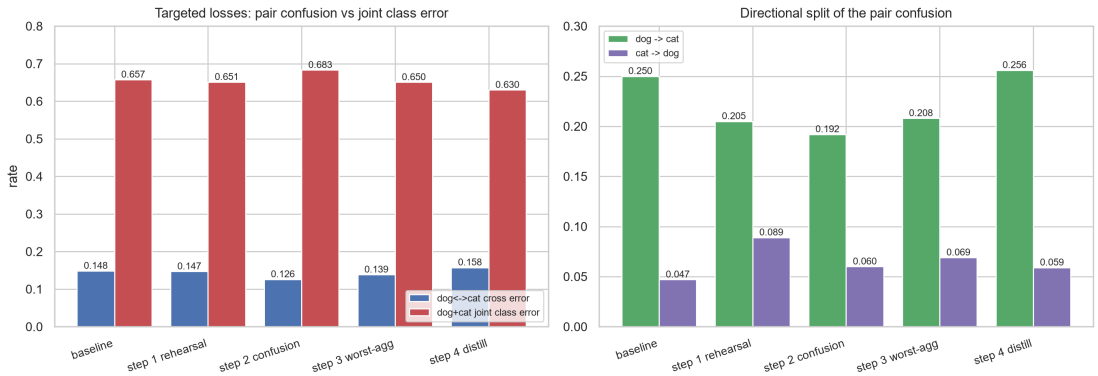
Step 4 weighted DEAL's KL term by exp(r_i(c)/T) with T=1.0. Dog recall rose to 0.282, frog to 0.706, and the gap fell to 0.424 (4.3%). The best dog architecture was EfficientNet-B1 before and after; its mean dog recall rose from 0.599 to 0.629, so the best-dog architecture did not change. This is the strongest mechanism-specific result, but still far from closure.

6.5 Full confusion matrices and the resolution probe

The per-class recalls above were supplemented with the full 10x10 confusion matrix of the global ensemble on the complete CIFAR-10 test set, for the baseline and all four rungs. The matrices make the Step 2 mechanism explicit: dog-to-cat fell 0.250 to 0.192, but dog-to-bird rose 0.188 to 0.213 and cat-to-bird 0.187 to 0.220. The pair loss did not remove error; it pushed the ambiguous animal mass into the neighbouring bird class. Dog's total error barely moved (0.749 to 0.730) and cat's worsened (0.566 to 0.636).

run	pair	a->b	b->a	cross	joint	cross share
baseline	dog-cat	0.2500	0.0470	0.1485	0.6575	0.2259
baseline	deer-horse	0.0900	0.0650	0.0775	0.5155	0.1503
baseline	automobile-truck	0.2660	0.1310	0.1985	0.3835	0.5176
STEP1_rehearsal	dog-cat	0.2050	0.0890	0.1470	0.6510	0.2258
STEP1_rehearsal	deer-horse	0.0710	0.0990	0.0850	0.4640	0.1832
STEP1_rehearsal	automobile-truck	0.1820	0.1910	0.1865	0.3760	0.4960
STEP2_confusion	dog-cat	0.1920	0.0600	0.1260	0.6830	0.1845
STEP2_confusion	deer-horse	0.0610	0.0850	0.0730	0.5030	0.1451
STEP2_confusion	automobile-truck	0.2430	0.1400	0.1915	0.3555	0.5387
STEP3_worst_agg	dog-cat	0.2080	0.0690	0.1385	0.6505	0.2129
STEP3_worst_agg	deer-horse	0.0870	0.0810	0.0840	0.4685	0.1793
STEP3_worst_agg	automobile-truck	0.2150	0.1580	0.1865	0.3710	0.5027
STEP4_distill	dog-cat	0.2560	0.0590	0.1575	0.6300	0.2500
STEP4_distill	deer-horse	0.0790	0.0740	0.0765	0.5140	0.1488
STEP4_distill	automobile-truck	0.2640	0.1320	0.1980	0.3835	0.5163

Cross uses only the pair's two off-diagonal cells; joint is the mean total error of the two classes, including errors to the other eight. Step 4 gives the lowest dog+cat joint error (0.6300) but its dog-cat cross error rises 6.1%. Step 1 halves automobile/truck directional asymmetry (0.135 to 0.009): a genuine rebalancing. Deer/horse is not a two-body pair; only 15% of its joint error is mutual, and deer-to-bird alone is 0.261.



Dog/cat error budget: pair confusion falls under Step 2 while joint class error rises.

A separate centralized probe (2,000 train / 500 test per class, one 3-conv CNN, 15 epochs, seed 0) measured the same pair at 32, 64 and 96 pixels. CIFAR-10 is natively 32x32, so 64 and 96 are upsampled and add no information. The probe degrades as resolution rises, and dog-cat cross-confusion stays flat (0.185 / 0.195 / 0.188).

resolution	AM	dog recall	cat recall	dog+cat cross	dog+cat joint
32	0.6950	0.5960	0.4940	0.1850	0.4550
64	0.6378	0.5560	0.4160	0.1950	0.5140
96	0.6092	0.5620	0.4000	0.1880	0.5190

The probe cannot test an information floor, because upsampling does not add pixels of information. What it does establish is that a centralized model on the same native 32x32 input reaches dog+cat joint error 0.455 against the federation's 0.6575. The federated dog/cat deficit is therefore not a missing-pixel problem. Whether the centralized residual 0.455 is resolution-limited or label-limited requires native 96x96 data (STL-10 downsampled to 64 and 32); the attempted download was abandoned incomplete.

7. What is interesting

- **1. A large class gap in a good client-FM run.** PHP-FL/linear has client-FM 0.086 and a class gap of 0.39 +/- 0.08 over three seeds (seed 0: 0.468, frog 69.4% vs dog 25.1%). A leaderboard sorted by FM would hide the failure.
- **2. FedAPEN makes the finding field-wide and is consistently worse.** Its gap is 0.60 +/- 0.09, above the pre-registered 30-point threshold in every seed, and it loses to PHP-FL in 6/6 per-seed comparisons. This is not a universal FL claim; it is field-wide across PHP-FL and FedAPEN.
- **3. The two axes order the runs differently, every seed.** The worst-gap cell is FedAPEN/linear in all three seeds and the best-client-FM cell in two of three. With three seeds this is a replicated ordering, not a correlation claim.
- **4. The pattern effect is method-dependent, not structural.** Entropy-minus-linear moves the gap by up to 15 points: worse for Standalone (3/3) and FedFV (3/3), mixed for PHP-FL, better for FedAPEN (3/3). Participation is not a neutral nuisance.
- **5. Four standard interventions do not close the gap.** Rehearsal, confusion loss, worst-class aggregation, and per-class distillation reduce it by at most 7.9% over one seed. The gap is not a simple optimization failure.
- **6. The confusion loss displaces error.** Dog-to-cat errors fall 23%, cat recall falls 16%, and the combined error rises. The 10x10 matrices show dog-to-bird 0.188 to 0.213 and cat-to-bird 0.187 to 0.220: the pair's error moved out of the pair rather than being removed. A centralized 32x32 probe reaches less than half the federated dog+cat joint error, so this is not an information floor at input resolution.
- **7. Standalone is tied with PHP-FL on the class axis.** Three seeds give 0.389 +/- 0.019 vs 0.387 +/- 0.077, with per-seed differences changing sign. Standalone pays 8.6 AM points and gains nothing on the class axis; the earlier gap/AM adjustment is dropped because there is no raw difference to adjust.

The most interesting single result is not the 39-point mean by itself. It is the combination of the rank disagreement, the seed instability, and the zero-sum confusion result: the class axis is a real, separate fairness object, its magnitude is seed-dependent, and one of the most natural ways to attack it moves errors rather than removing them.

8. Limitations and what is not established

- **No Table 2 reproduction.** The original 66.94/8.07 baseline assertion was not executed after the scope change. This repository does not claim to reproduce the paper's full CIFAR-10 numbers.
- **Half-scale only.** 10 clients, 60 rounds, four CNNs, one local epoch, 2,500/500 per client. Absolute AM/FM are not comparable to the paper. The three-seed sweep changes the uncertainty, not the scale.
- **Three seeds.** Enough to expose seed noise (sd up to 0.085) but not enough for significance on the FedFV-vs-PHPFL margin. Pairwise claims are sign counts, not tests.
- **Four methods, not a field survey.** The field-wide scope covers two MH-FL methods (PHP-FL and FedAPEN). FedFV is measured but is a reimplementation applied to the shared model; FedHEAL, FedAWE, and FedAU

are not implemented.

- **Support sparsity.** Per-client class cells are sparse. The 10-sample threshold is a pre-declared compromise; the global class headline uses the full test set, but the client-cell tail is still noisy.
- **Descriptive, not causal.** The analysis does not identify why dog/cat/deer/horse are weak. The confusion audit locates the errors (one animal cluster plus automobile/truck) but does not establish the mechanism. The label-ambiguity interpretation of the residual centralized error is untested, because the CIFAR 64/96 probe is upsampled.

9. Future directions

The project is now a measurement paper. The most valuable next step is not another method; it is to test the mechanism behind the zero-sum result and to make the reporting practice standard.

9.1 Completed: resolution probe and pairwise error budget

The resolution sweep and the pairwise budget are now done and reported in Section 6.5. The pair budget shows that targeted losses displace error more often than they remove it, and that deer/horse is not primarily a pair problem at all. The CIFAR probe cannot settle the information-floor question because 32x32 data cannot be upsampled into new information, and the probe degraded as resolution rose.

The remaining primary experiment is a native high-resolution control: STL-10, natively 96x96, downsampled to 64 and 32 so the same scenes are presented at three genuine information levels. Only then can a resolution-limited floor be separated from label ambiguity. A label audit of dog/cat/deer/bird images is the fallback if the native control still shows flat pair confusion.

9.2 Zero-training: architecture decomposition

The committed per-client logs already contain architecture labels and per-class recalls. A zero-training analysis can test whether the class gap is concentrated in specific architectures. EfficientNet-B1 was the best dog architecture (mean dog recall 0.599 baseline, 0.629 in Step 4); if other architectures contribute most of the cat/dog errors, the gap is partly an inductive-bias problem rather than federation.

9.3 More seeds on the FedFV margin

Three seeds established the seed noise (sd up to 0.085) but cannot separate FedFV/linear's 3.4-point mean advantage over PHP-FL. With that sd, n >= 10 seeds is the smallest budget that could distinguish them; it is the highest-value run left. A full-scale CIFAR-10 audit (20 clients, 3 local epochs, multiple seeds) remains the prerequisite for any universal claim.

9.4 More MH-FL methods

FedFV is now measured (as a shared-model port). FedHEAL, FedAWE, and FedAU are not. Identifying the exact FedHEAL paper PHP-FL cites is a prerequisite, because several methods share the name. Porting the rest into the same heterogeneous setting would test whether class-FM is a general problem or specific to PHP-FL/FedAPEN. This is measurement, not a new method.

9.5 Reporting standard

The practical recommendation is a three-number report plus the matrix: client-FM, class-FM, worst-class recall, and the client-by-class matrix or a link to it. class_FM should be reported, not optimized. A model that wins client-FM but loses class-FM should not be presented as unambiguously fairer.

10. Reproducibility and artifact map

The public repository contains the paper source, analysis code, result CSVs, structured per-run logs, and figures. Large artifacts are excluded.

Path	Contents
phofl/paper/	Abstract through conclusion (sections 00-07)
phofl/analysis/	Client x class matrices, support filtering, AM normalization, ranks, 10x10 confusion matrices, pair error budgets, resolution probe, three-seed sweep
phofl_method/	Ladder per-step CSVs, findings, LADDER_SUMMARY.md
phofl_half/results/	Half-scale result CSVs, including the 24-row seed sweep
phofl_half/logs/	Per-run round metrics and histories
phofl/results/, phofl/logs/	Earlier A/B/C scoped results
phofl/figures/	300 DPI figures
phofl_experiments.ipynb	Resumable A/B/C notebook
phofl_lab.py	Runtime harness, incl. FedFV aggregation
analyze_two_axis.py	Rebuilds the 2-D analysis
analyze_confusion.py	Builds the 10x10 matrices and pair budget
resolution_probe.py	Centralized 32/64/96 probe
run_seed_sweep.py	Three-method + FedFV seed sweep
run_step1.py ... run_step4.py	Diagnostic ladder
tests/test_fedfv.py	FedFV aggregation unit tests

Excluded: third_party clones, generated CIFAR-10 partitions, checkpoints, run_store, phofl_pinned, and .venv-pinned. DATA.md documents regeneration. The repository is at github.com/Paul4805/phofl-class-fairness; release v0.1.0 is tagged.

10.1 Key commands

Rebuild the committed analysis without training:

```
python analyze_two_axis.py
python analyze_confusion.py
python analyze_seed_sweep.py
python make_final_table.py
python -m pytest -q
```

One-line conclusion

Client-FM is a useful but incomplete fairness report. The class axis is a real second dimension, but its magnitude is seed-dependent (0.39 +/- 0.08 for PHP-FL/linear, not a stable 44 points), it ranks methods differently in every seed, and four standard interventions do not close it. Only FedAPEN's larger gap is established across seeds. Report class_FM and worst-class recall alongside client-FM with seed spread, and do not treat class_FM as an objective.

References

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